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RESEARCH-ARTICLE

Interpretable Knowledge Tracing with Multiscale State Representation

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ABSTRACT

Knowledge Tracing (KT) is vital for education, continuously monitoring students' knowledge states (mastery of knowledge) as they interact with online education materials. Despite significant advancements in deep learning-based KT models, existing approaches often struggle to strike the right balance in granularity, leading to either overly coarse or excessively fine tracing and representation of students' knowledge states, thereby limiting their performance. Additionally, achieving a high-performing model while ensuring interpretability presents a challenge. Therefore, in this paper, we propose a novel approach called Multiscale-state-based Interpretable Knowledge Tracing (MIKT). Specifically, MIKT traces students' knowledge states on two scales: a coarse-grained representation to trace students' domain knowledge state, and a fine-grained representation to monitor their conceptual knowledge state. Furthermore, the classical psychological measurement model, IRT (Item Response Theory), is introduced to explain the prediction process of MIKT, enhancing its interpretability without sacrificing performance. Additionally, we extended the Rasch representation method to effectively handle scenarios where questions are associated with multiple concepts, making it more applicable to real-world situations. We extensively compared MIKT with 20 state-of-the-art KT models on four widely-used public datasets. Experimental results demonstrate that MIKT outperforms other models while maintaining its interpretability. Moreover, experimental observations have revealed that our proposed extended Rasch representation method not only benefits MIKT but also significantly improves the performance of other KT baseline models. The code can be found at <https://github.com/lilstrawberry/MIKT>.

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CCS CONCEPTS

• Applied computing → Education; Distance learning.

KEYWORDS

knowledge tracing, knowledge state representation, interpretable, educational data mining

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1 INTRODUCTION

In the context of the web, where online education platforms like Coursera, MOOC, and others have become increasingly prevalent, Knowledge Tracing (KT) emerges as a critical topic. With the vast array of educational resources available online, it has become a crucial task for online learning platforms to model the learning process of their users, and further provide their users a personalized learning guidance as they navigate through web-based learning environments[46]. KT plays a pivotal role in achieving this by continuously monitoring and assessing students' knowledge states based on their interactions with online educational materials[3].

Recently, models based on deep learning have shown outstanding performance[10, 18, 25, 47]. Given that KT's central task is to trace students' knowledge states, designing an effective encoding for representing knowledge states becomes of paramount importance. Additionally, deep learning-based KT models often lack interpretability, which undoubtedly impedes their further application.

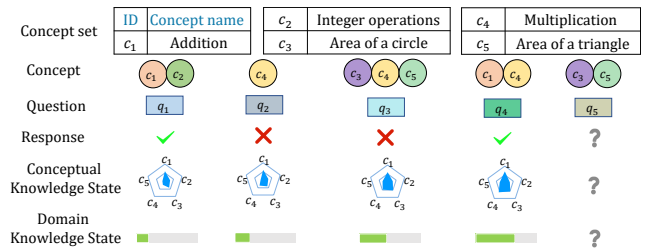


Figure 1: A simple example about knowledge tracing

We have presented a simple example of knowledge tracing in Figure 1, where each question is associated with one or more concepts. Students practice on different questions, gradually increasing their knowledge during the learning process. To represent the student's knowledge state, the graph-based KT model[21, 36] traces the student's knowledge states on all concepts (referred to as conceptual knowledge state in this paper) using hidden vectors and propagates these knowledge states through the specific graph structure. On the other hand, the sequential-based KT models[11, 27] do not trace the student's knowledge state on each concept. Instead, they directly model the student's current knowledge state (referred to as domain knowledge state in this paper) based on the student's historical interactions and represent this knowledge state using a hidden vector. These two modeling approaches are vastly different but have both achieved impressive success. It inevitably leads us to wonder: can we combine these two methods to better trace students' knowledge state? From a modeling perspective, the graph-based KT model can finely trace a student's knowledge state on the current question, while the sequential-based KT model provides a more coarse-grained representation of the student's overall knowledge state. We believe that both coarse-grained and fine-grained tracing of students' knowledge state is necessary. For example, suppose a student has recently answered multiple questions incorrectly. Now, when the student encounters a question that they are actually good at, the graph-based KT model might predict that the student will answer correctly because it assumes the student has a strong grasp of the underlying concept. On the other hand, the sequential-based KT model might predict that the student will answer incorrectly due to the student's recent history of consistently wrong responses. It might perceive that the student's learning state has been poor lately, and therefore, even when the student encounters a question they should know, the model may still predict an incorrect response. Integrating both modeling approaches could potentially lead to a more comprehensive and accurate KT model. By leveraging the strengths of both fine-grained and coarse-grained tracing, we can better understand and predict students' learning progress.

Therefore, we believe that combining the two mentioned modeling approaches is a promising idea. However, several challenges need to be addressed. First, to the best of our knowledge, we are the first to propose the integration of graph-based KT models and sequential-based KT models. Effectively combining these two approaches to leverage their respective strengths remains an unexplored endeavor. How can we integrate these two modeling approaches to harness their potential? Second, sparse interactions between students and questions make it difficult for the model to accurately trace the students' knowledge states[31]. To address this issue, we need to consider how to represent the questions. This is a common problem faced by most KT models. Although AKT[10] introduced the Rasch[26] representation method, which significantly improved model performance and benefited other KT models, AKT assumes that a question is only related to a single concept, which does not align with real-world scenarios. Lastly, deep learning-based KT models often lack interpretability. The opaque decision-making process of these models limits their further applications. On the other hand, KT models that consider interpretability typically sacrifice performance, creating a trade-off between high

performance and interpretability[5]. Therefore, designing a high-performance model with interpretability is a critical challenge.

In this paper, we propose a novel approach called Multiscale-state-based Interpretable Knowledge Tracing (MIKT) to address the challenges mentioned above. Specifically, MIKT trace students' knowledge states on two scales: a coarse-grained representation to monitor students' domain knowledge state and a fine-grained representation to monitor their conceptual knowledge state. The domain knowledge state is constructed based on the student's entire historical interactions, taking into account forgetting behavior at each time step. The conceptual knowledge state is built from the student's answers to questions related to specific concepts, considering the student's domain knowledge state and the time interval since their last responses for each concept to account for forgetting behavior. Since each question involves multiple concepts, an attention mechanism is used to aggregate conceptual knowledge states with different weights. When combining these two knowledge states, MIKT utilizes an attention mechanism to differentiate their respective roles. Additionally, MIKT extends the Rasch model to handle questions covering multiple concepts, aggregating them using attention while considering their difficulty levels. To ensure interpretability, MIKT not only relies on attention mechanisms but also introduces an IRT prediction module to explain model predictions. MIKT fits well with the IRT[40] prediction module since it can derive student abilities from multiple states and question difficulty using the expanded Rasch module, both of which are required by the IRT prediction module. This enhances the model's interpretability while maintaining performance. We extensively compare MIKT with 20 state-of-the-art KT models on four widely-used public datasets. Experimental results demonstrate that MIKT exhibits excellent performance and high interpretability.

This paper proposes a novel model called Multiscale-state-based Interpretable Knowledge Tracing (MIKT), and its contributions can be summarized as follows:

- To the best of our knowledge, we are the first to consider combining students' knowledge states from multiple scales, providing a new perspective for better tracing students' knowledge states.
- We fully utilize attention mechanisms to facilitate the transparent modeling process of MIKT. Additionally, an IRT prediction module is designed to interpret MIKT's output, seamlessly complementing MIKT and enhancing interpretability without compromising performance.
- We extend the Rasch model, which not only benefits MIKT but also significantly improves the performance of other KT baseline models.
- Through experiments on four widely-used public datasets and comparison with 20 state-of-the-art KT models, the results demonstrate that MIKT exhibits excellent performance in predicting student performance. Additionally, interpretability experiments confirm the high level of explainability achieved by MIKT.

2 RELATED WORK

In this section, we first review the modeling approach of knowledge tracing and then introduce interpretable knowledge tracing.

2.1 Knowledge State Modeling for Deep Knowledge Tracing

Based on existing knowledge tracing models, we classify the mechanisms for representing students' knowledge states into four distinct categories:

- **State-Sequential:** Models like DKT[25], ATKT[11], DIMKT[27], etc., are representative sequential-based KT models that trace students' knowledge states at discrete time steps.
- **State-Latent:** Represented by models such as DKVMN[47], SKVMN[1], DGMN[2], etc., these KT models assume that all questions are related to their predefined latent concepts and trace students' knowledge states on these latent concepts.
- **State-Graph:** Models like GKT[21], SKT[36], HGKT[35], etc., fall into this category. They are graph-based KT models that trace students' knowledge states on individual concepts and propagate them within the graph.
- **State-Free:** SAKT[23], AKT[10], DTransformer[46], sparseKT[12], etc., are leading attention-based KT models in this category. They do not trace students' knowledge states at each discrete time step; instead, they dynamically construct the students' current knowledge states based on their historical interactions.

2.2 Interpretable Deep Knowledge Tracing

Recently, an increasing number of interpretable methods have been adopted in KT models. These models can be broadly categorized into three types:

- **Post-hoc local explanation:** Aims to explain why the model makes certain predictions or decisions, such as [19] using LRP[4] techniques to propagate relevance scores from the model's output layer to the input layer for interpreting KT models.
- **Globally interpretable with an interpretable structure:** Aims to design an interpretable model structure to understand the process of modeling knowledge states. For instance, [38] embeds an interpretable cognitive framework in the model to understand students' knowledge state modeling process.
- **Globally interpretable with interpretable parameters:** Aims to directly utilize interpretable parameters derived from the model to explain its predictions. For example, QIKT[5] exports three interpretable ability scores of students and uses them to make predictions in the output layer.

From a modeling perspective, MIKT (state-Sequential + state-graph) can provide better interpretability than DKT (sequence-based) (because MIKT can trace students' state of each concept), MIKT (state-Sequential + state-graph) can provide a more comprehensive knowledge state than GKT (state-graph), and the domain knowledge state significantly enhances the performance of the model, etc. From an interpretable perspective, MIKT offers several advantages compared to interpretable methods such as Deep-IRT and QIKT. MIKT can provide more comprehensive explanatory

parameters instead of being limited to predictions, which is a disadvantage of Deep-IRT and QIKT. Additionally, the calculation of each interpretable parameter in MIKT is transparent.

3 MIKT FRAMEWORK

In this chapter, we will provide a detailed introduction to MIKT, and its overall architecture is illustrated in Figure 2. First, we present the problem formulation of KT. Subsequently, we provide a comprehensive overview of each module in MIKT: the Expanded Rasch Module, the Cognitive Thinking Module, the IRT Prediction Module, and the Cognitive Update Module. Finally, we outline the training process of the model.

3.1 Problem Formulation

The KT task can be formulated as follows: Given a student's historical interaction sequence represented as $X = (q_1, Neighbor_{q_1}, a_1), (q_2, Neighbor_{q_2}, a_2), \dots, (q_t, Neighbor_{q_t}, a_t)$, where q_t denotes the question answered by the student at time t , $Neighbor_{q_t}$ represents the set of related concepts to q_t and $a_t \in \{0, 1\}$ indicates whether the student's response to the question is incorrect ($a_t = 0$) or correct ($a_t = 1$). The task of KT is to predict the probability $p(a_{t+1} = 1 | X, q_{t+1})$ of a student answering the next question q_{t+1} correctly.

3.2 Expanded Rasch Module

Our proposed Expanded Rasch Module is capable of handling scenarios where a question is associated with multiple concepts. Specifically, given the original question embedding matrix $K \in R^{n \times d}$ and the concept embedding matrix $C \in R^{m \times d}$, where n denotes the total number of questions, d represents the embedding dimension, and m indicates the total number of concepts, we use K_i to represent the i -th row of K and C_i to represent the i -th row of C . For the question q_t at the current moment, in order to distinguish the effects of different concepts on the question, the concept representation MC_{q_t} contained in the question q_t is computed as follows:

$$MC_{q_t} = \sum_{j \in Neighbor_{q_t}} qa_j * C_j$$

$$qa_j = \frac{\exp(\frac{K_{q_t}^T C_j}{\sqrt{d}})}{\sum_{i \in Neighbor_{q_t}} \exp(\frac{K_{q_t}^T C_i}{\sqrt{d}})} \quad (1)$$

Among these, $Neighbor_{q_t}$ represents the set of concepts related to question q_t , and qa_j can be regarded as the degree of relevance between the question and its associated concepts. Furthermore, inspired by Rasch[26] theory, we should pay attention to the characteristics inherent in the question itself, such as its difficulty level and how it differs from the associated concepts. Specifically, the features OF_{q_t} encompassed in the question q_t are obtained through the following equation:

$$OF_{q_t} = diff_{q_t} * (W_1 \sum_{j \in Neighbor_{q_t}} \frac{1}{|Neighbor_{q_t}|} C_j + b_1) \quad (2)$$

Among these, $diff_{q_t}$ represents the difficulty level of question q_t , which is a learnable parameter. W_1 and b_1 are learnable parameters, and $|Neighbor_{q_t}|$ denotes the total number of concepts associated

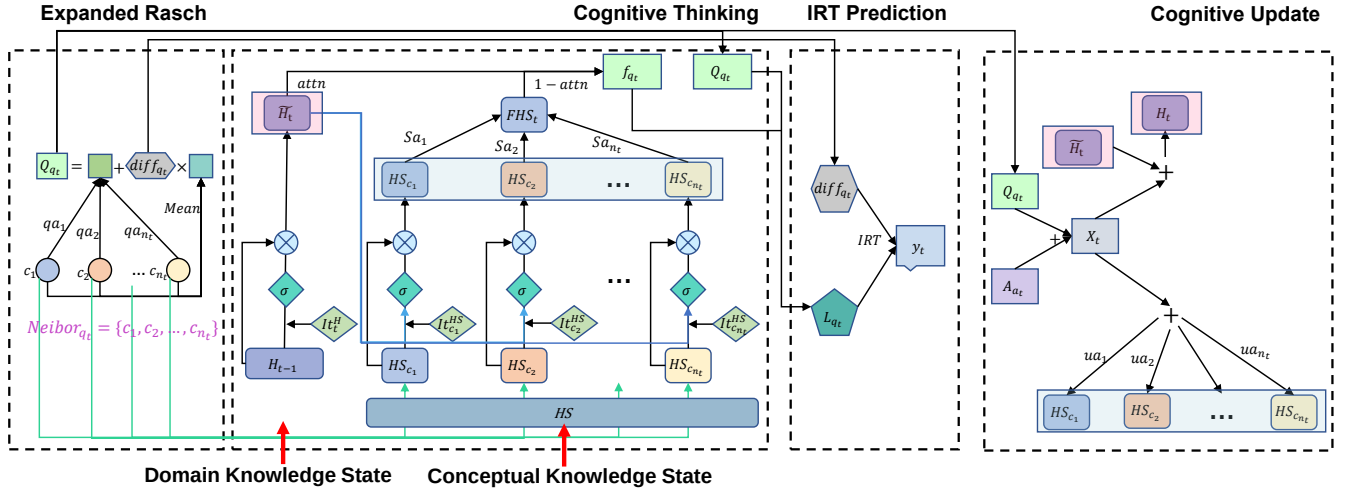


Figure 2: The overall framework of MIKT. Firstly, the final question embeddings are obtained by utilizing the Expanded Rasch module. In the Cognitive Thinking module, the forgotten representations of Domain Knowledge State and Conceptual Knowledge State are obtained, respectively, and then aggregated. Predictions are made using the IRT Prediction module. In the Cognitive Update module, the Domain Knowledge State and Conceptual Knowledge State used in the Cognitive Thinking module are updated based on the student's responses, respectively.

with question q_t . Note that the calculation here takes the average of related concepts, mainly because we thought the degree of change of the question is compared to the concepts it is associated with.

Finally, assuming the final question embedding matrix is represented as $Q \in R^{n \times d}$, where Q_i denotes the i -th row of Q , then:

$$Q_{q_t} = MC_{q_t} + OF_{q_t} \quad (3)$$

3.3 Cognitive Thinking Module

MIKT traces students' domain knowledge state $H \in R^{T \times d}$ and conceptual knowledge state $HS \in R^{m \times d}$, where T represents the total number of time steps. H_i and HS_i represent the i -th row of H and HS , respectively. At the time step t :

We denote the student's current domain knowledge state as $\tilde{H}_t$. It is essential to note that the student's knowledge state is not constant but gradually fades over time, considering the forgetting behavior. For the student's domain knowledge state at the previous time step H_{t-1} , if the time interval since the last interaction is represented as It_t^H after embedding, the domain knowledge state of the student after forgetting can be calculated as follows:

$$\tilde{H}_t = H_{t-1} * \sigma(\beta(H_{t-1} \oplus It_t^H)) \quad (4)$$

Among them, σ represents the Sigmoid function, β represents non-linear transformation, and $\oplus$ represents concatenation operation.

For the concept set $Neibor_{q_t}$ related to the current question q_t , MIKT trace students' knowledge states on these concepts. Specifically, for any concept $j \in Neibor_{q_t}$, MIKT obtain the student's knowledge state on concept j denoted as HS_j based on its index. MIKT also consider the forgetting behavior for the conceptual knowledge state, and denote the time interval since the student's last response to concept j as It_j^{HS} , which is represented through embedding. Thus, the student's conceptual knowledge state is obtained

after considering forgetting behavior:

$$HS_j = HS_j * \sigma(\xi(HS_j \oplus It_j^{HS} \oplus \tilde{H}_t)) \quad (5)$$

Please note that the clarification here is that the time interval in this context is the interval in terms of time steps. For instance, if a student's last response for concept j was at time step α , then the time interval from the current moment is $t - \alpha$. This approach offers two advantages: firstly, it simplifies calculations, and secondly, it eliminates the need to introduce additional time-related features, which may not be present in all datasets. Furthermore, the non-linear transformation is represented by ξ . Different notations for non-linear transformations represent different learnable parameters. It is worth noting that during the computation of HS_j , we incorporate $\tilde{H}_t$. This consideration arises from acknowledging that a student's domain knowledge state may impact their conceptual knowledge state. Consequently, MIKT aggregate the student's knowledge states for all relevant concepts to derive the student's current conceptual knowledge state at time t , denoted as FHS_t :

$$FHS_t = \sum_{j \in Neibor_{q_t}} Sa_j * HS_j$$

$$Sa_j = \frac{\exp(\frac{Q_{q_t}^T HS_j}{\sqrt{d}})}{\sum_{i \in Neibor_{q_t}} \exp(\frac{Q_{q_t}^T HS_i}{\sqrt{d}})} \quad (6)$$

Sa_j can be understood as the degree of relevance between the question and HS_j . In order to derive the student's ultimate knowledge state f_{q_t} , we employ varying levels of attention to $\tilde{H}_t$ and FHS_t :

$$f_{q_t} = attn * \tilde{H}_t \oplus (1 - attn) * FHS_t$$

$$attn = \sigma(\pi(Q_{q_t} \oplus \tilde{H}_t \oplus FHS_t)) \quad (7)$$

Where π represents non-linear transformation, $attn$ is an intermediate variable that represents the student's attention to the domain knowledge state $\bar{H}_t$. Clearly, $1 - attn$ indicates the student's attention to the conceptual knowledge state FHS_t . It's worth noting that concatenation is used here instead of addition, considering the different roles played by various levels of knowledge states.

3.4 IRT Prediction Module

In order to increase the transparency of MIKT's final decision-making process, an IRT prediction module has been implemented to provide explanations for MIKT's predictions. This module utilizes the student's current knowledge state f_{q_t} and the question Q_{q_t} to calculate the student's ability value L_{q_t} in relation to the specific question:

$$L_{q_t} = \sigma(\varphi(f_{q_t} \oplus Q_{q_t})) \quad (8)$$

φ represents non-linear transformation. Afterwards, we extract the current question's difficulty value $diff_{q_t}$ from the expanded Rasch module. Following the principles of the IRT[40] method, a student's prediction is influenced by their ability value and the question's difficulty value (more details on IRT are provided in Appendix A). MIKT's prediction y_t is determined through the following equation:

$$y_t = \sigma(5 * (L_{q_t} - diff_{q_t})) \quad (9)$$

Please be aware that the multiplication by 5 serves the main purpose of achieving smoother output values for the Sigmoid function and does not hold any other particular significance.

3.5 Cognitive Update Module

After students complete their answers, the knowledge will be updated according to their performance, and the acquired knowledge will be represented as X_t :

$$X_t = Q_{q_t} + A_{a_t} \quad (10)$$

Here, $A \in R^{2 \times d}$ denotes the answer embedding, and A_i represent the i -th row of A . We use the following equation to obtain the student's final domain knowledge state H_t at the moment t after the process of learning and acquisition:

$$H_t = \widetilde{H}_t + \text{Tanh}(\psi(X_t)) \quad (11)$$

The Tanh function is denoted as Tanh , and ψ represents non-linear transformation. The gain in conceptual knowledge state be represented as:

$$G_t = \text{Tanh}(\omega(X_t)) \quad (12)$$

Here, ω denotes non-linear transformation. For the knowledge state of any relevant concept $j \in \text{Neighbor}_{q_t}$ at the current time step, it is updated using the following equation:

$$HS_j = HS_j + ua_j * G_t$$

$$ua_j = \frac{\exp(\frac{G_t^T HS_j}{\sqrt{d}})}{\sum_{i \in \text{Neighbor}_{q_t}} \exp(\frac{G_t^T HS_i}{\sqrt{d}})} \quad (13)$$

ua_j can be understood as how much knowledge the student's HS_j has gained from G_t .

3.6 Model Training

The KT loss $Loss_{KT}$ is defined as the binary cross-entropy loss between the prediction y_t and the true answer a_t , and it is computed as follows:

$$Loss_{KT} = - \sum_{i=1}^T (a_i \log y_t + (1 - a_i) \log(1 - y_t)) \quad (14)$$

We utilize the Adam[13] algorithm to optimize the model.

4 EXPERIMENT

In this section, we conducted extensive experiments to answer the following questions:

- **RQ1:** How does MIKT perform?
- **RQ2:** How do different components of MIKT impact its performance? Is the proposed multiscale-state tracing beneficial?
- **RQ3:** How does MIKT provide explanations for its predictions?
- **RQ4:** Can the expanded Rasch representation method proposed by MIKT improve the performance of other knowledge tracing baseline models?

4.1 Experimental Setting

4.1.1 Datasets. We evaluated the performance of MIKT on four commonly used public datasets: ASSIST09¹, ASSIST12², EdNet³ and Eedi⁴. The specific introduction and processing methods of the dataset can be found in Appendix B, and the statistical information of the dataset is presented in Table 1.

Table 1: Summary statistics of processed datasets.

	ASSIST09	ASSIST12	EdNet	Eedi
# Student	4,160	5,000	5,000	5,000
# Question	15,680	36,056	11,775	26,706
# Concept	167	242	1,837	1,050
# Interaction	207,659	717,188	1,156,254	597,124

4.1.2 Baseline Model. To assess the performance of MIKT, we compared it with 20 state-of-the-art KT models as follows: DKT[25], DKT+[45], KQN[14], DKT+forgetting[20], PEBG+DKT[16], GIKT[43], ATKT[11], CDKT[8], DIMKT[27], QIKT[5], AT-DKT[17], DKVMN[47], Deep-IRT[44], GKT[21], SAKT[23], SAINT[6], AKT[10], CL4KT[15], simpleKT[18], DTransformer[46]. Their details are in Appendix C.

4.1.3 Implementation Details. We implemented MIKT using PyTorch[24] with the following settings: the learning rate was set to 0.002, the batch size was 80, and the embedding dimension was 64. Additionally, L2 weight regularization with a weight decay of 1e-5 was applied to the model's weights. To avoid the issue of gradient explosion, we consistently set the gradient clipping threshold to

¹<https://sites.google.com/site/assistentmentsdata/home/2009-2010-assistentment-data>

²<https://sites.google.com/site/assistentmentsdata/home/2012-13-school-data-with-affect>

³<https://github.com/rriid/ednet>

⁴<https://eedi.com/projects/neurips-education-challenge>

Table 2: Comparison of MIKT and 20 KT models on four datasets. Best results in bold, next best underlined. * indicates t-test p-value < 0.05 compared to the second best result. Model grouping details can be found in Section 2 (RELATED WORK).

Model	Model Group	Interpretable	ASSIST09		ASSIST12		EdNet		Eedi	
			AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC
DKT[25]	State-Sequential	×	0.7684	0.7297	0.7328	0.7367	0.7006	0.7129	0.7629	0.7182
DKT+[45]	State-Sequential	×	0.7783	0.7337	0.7373	0.7350	0.7028	0.6698	0.7484	0.7079
KQN[14]	State-Sequential	✓	0.7546	0.7249	0.7230	0.7330	0.6909	0.7117	0.7583	0.7143
DKT+forgetting[20]	State-Sequential	×	0.7717	0.7295	0.7362	0.7359	0.7018	0.7159	0.7642	0.7186
PEBG+DKT[16]	State-Sequential	×	0.7738	0.7329	0.7518	0.7495	0.7571	0.7366	0.7853	0.7310
GIKT[43]	State-Sequential	×	0.7726	0.7301	0.7672	0.7506	0.7640	0.7366	0.7924	0.7362
ATKT[11]	State-Sequential	×	0.7735	0.7332	0.7347	0.7363	0.7027	0.7109	0.7663	0.7195
CDKT[8]	State-Sequential	×	0.7733	0.7297	0.7720	0.7547	0.7645	<u>0.7386</u>	0.7920	0.7360
DIMKT[27]	State-Sequential	×	0.7704	0.7310	0.7621	0.7484	0.7623	0.7368	0.7908	0.7338
QIKT[5]	State-Sequential	✓	0.7801	0.7377	0.7707	0.7529	0.7579	0.7327	<u>0.7932</u>	<u>0.7363</u>
AT-DKT[17]	State-Sequential	×	0.7671	0.7293	0.7425	0.7405	0.7039	0.7136	0.7649	0.7180
DKVMN[47]	State-Latent	×	0.7629	0.7266	0.7228	0.7329	0.6975	0.7120	0.7590	0.7162
Deep-IRT[44]	State-Latent	✓	0.7657	0.7279	0.7253	0.7345	0.6997	0.7124	0.7609	0.7173
GKT[21]	State-Graph	×	0.7666	0.7290	0.7261	0.7333	0.6943	0.7104	0.7618	0.7170
SAKT[23]	State-Free	×	0.7564	0.7192	0.7296	0.7348	0.6956	0.7115	0.7556	0.7123
SAINT[6]	State-Free	×	0.7515	0.7134	0.7643	0.7477	0.7621	0.7370	0.7866	0.7293
AKT[10]	State-Free	×	<u>0.7850</u>	<u>0.7429</u>	<u>0.7830</u>	<u>0.7599</u>	<u>0.7647</u>	0.7385	0.7882	0.7340
CL4KT[15]	State-Free	×	0.7626	0.7275	0.7236	0.7331	0.6965	0.7118	0.7583	0.7147
simpleKT[18]	State-Free	×	0.7772	0.7315	0.7786	0.7571	0.7627	0.7373	0.7885	0.7307
DTransformer[46]	State-Free	×	0.7646	0.7223	0.7672	0.7515	0.7501	0.6954	0.7531	0.7315
MIKT	State-Sequential+State-Graph	✓	0.7938*	0.7454*	0.7834	0.7608	0.7703*	0.7430*	0.7954*	0.7392*

15.0 during training. For data preprocessing, we removed sequences with a length less than 3 from the dataset. Since the input sequence lengths varied, we uniformly set all sequences to a fixed length of 200. For each dataset, we used 80% of all sequences as the training set and 20% as the test set[43, 46]. We conducted the experiment five times and reported the average results[2, 32, 33]. All models are trained on a Linux server with two 2.00GHz Intel(R) Xeon(R) CPUs and a Nvidia Tesla P100-PCIE-16GB GPU. Consistent with prior work[22, 29, 30, 34, 41], we use AUC (Area Under the Curve) as the first evaluation metric and ACC (Accuracy) as the second metric. The greater their values, the better the model's performance.

4.2 Performance (RQ1)

4.2.1 Overall Performance. Table 2 presents the performance comparison between MIKT and other KT models, with the best results highlighted in bold and the second-best results underlined. According to Table 2, we can observe the following: (1) MIKT demonstrates superior performance across all datasets, underscoring the effectiveness of the proposed method in this paper. (2) When comparing MIKT to State-Sequential and State-Graph KT models, MIKT exhibits a noticeable performance improvement, indicating the effectiveness of constructing students' knowledge states from multiple perspectives. (3) Comparing Deep-IRT to DKVMN in State-Latent, Deep-IRT enhances model interpretability through the introduction of IRT methods while slightly improving performance, suggesting that IRT methods do not significantly sacrifice KT model performance. (4) Among the State-Free KT models, AKT stands out as a robust baseline, owing to its two key components: representing questions using the Rasch method and employing a context-aware attention structure to mimic student forgetting behavior. This underscores the importance of question representation and forgetting in KT. (5) Among all compared baseline models, none achieve both

high performance and interpretability. In contrast, MIKT maintains high performance while also offering interpretability.

4.2.2 $T + N$ Prediction. To better simulate real student question-answering scenarios and evaluate the stability of MIKT performance, we conducted a $T + N$ prediction experiment. Specifically, in this experiment, we not only predicted the performance of students in answering questions at the next moment $T + 1$ (assuming the current moment is T) but also predicted their performance at moments $T + 2, T + 3, \dots, T + N$. As shown in Figures 3 and 6 (in Appendix), we compared MIKT with some high-performing KT models (see Table 2) on all datasets and made the following observations: (1) As the value of N in $T + N$ increased, the performance of all models generally exhibited a decreasing trend, which is expected because KT models primarily predict performance at $T + 1$. However, in all cases, MIKT showed the slowest decrease in performance. This demonstrates the advantage of MIKT, namely, its ability to trace students' knowledge states across multiple scales, which proves to be more stable compared to tracing knowledge states at a single scale. (2) We found that AKT exhibited unstable predictive performance in this experiment. This may be because State-Free KT methods construct knowledge states based on specific questions as queries, and when there is a need to predict multiple different questions, these knowledge states may not be highly relevant to the questions that need to be answered. (3) MIKT consistently maintained high performance in all situations and consistently outperformed other KT models. This indicates that MIKT's ability to trace knowledge states is both stable and effective.

4.3 Ablation Studies (RQ2)

Four variants of MIKT were constructed to explore the impact of different components on MIKT, as shown in Table 3. Specifically, "w/o ERM" removes the Expanded Rasch Module, "w/o Forget"

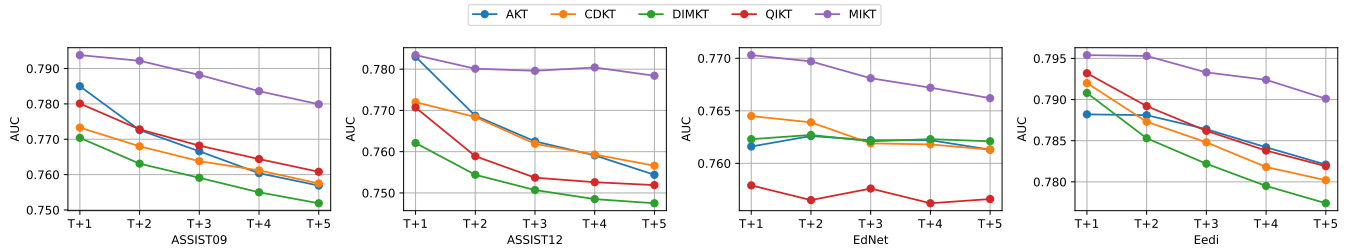


Figure 3: Comparing T+N prediction performance (AUC) of MIKT and well-performing KT models across four datasets.

Table 3: Performance comparison of ablation study.

Model	ASSIST09		ASSIST12		EdNet		Eedi	
	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC
MIKT w/o ERM	0.7822	0.7371	0.7787	0.7577	0.7669	0.7415	0.7925	0.7378
MIKT w/o Forget	0.7780	0.7314	0.7657	0.7504	0.7649	0.7392	0.7829	0.7296
MIKT w/o DKS	0.7723	0.7280	0.7658	0.7498	0.7535	0.7322	0.7488	0.7076
MIKT w/o CKS	0.7868	0.7412	0.7756	0.7571	0.7658	0.7395	0.7924	0.7368
MIKT	0.7938	0.7454	0.7834	0.7608	0.7703	0.7430	0.7954	0.7392

disregards the Forgetting of knowledge states, "w/o DKS" eliminates Domain Knowledge States, and "w/o CKS" omits Conceptual Knowledge States. The following observations were made: (1) "w/o ERM" exhibited a similar degree of performance decline across all datasets, with the most noticeable decrease observed in ASSIST09. This underscores the importance of question representation. It's worth noting that, in comparison to other modules, the reduction in performance with "w/o ERM" is not particularly pronounced (Detailed data can be found in Table 6). This may, to some extent, indicate the robustness of MIKT in tracing students' knowledge states, making it less reliant on question representations. In subsequent experiments, we will demonstrate how ERM-based question representation significantly enhances the performance of some simpler KT models. (2) "w/o Forget" experienced an average performance decrease of around 1%, indicating the significant role of forgetting in the student's knowledge state. This aligns with intuition, as knowledge forgetting is an essential aspect of the learning process. (3) "w/o DKS" and "w/o CKS" both exhibited varying degrees of performance decrease, with "w/o DKS" showing the most significant decline. This is in line with expectations since building "DKS" involves the entire interaction history of the student, whereas constructing "CKS" only relies on relevant interaction history. Furthermore, the combination of both "DKS" and "CKS" significantly improved their respective performance, underscoring the effectiveness of the multi-scale approach proposed in this paper.

4.4 Explainable thinking process (RQ3)

To demonstrate the interpretability of MIKT, we randomly selected a student from the ASSIST09 dataset and observed MIKT's prediction process for the student's response to question 4457 and its related concepts 33, 53, and 85, as shown in Figure 4. First, in the Expanded Rasch module, MIKT estimated the relevance of question 4457 to these concepts as 0.3310, 0.3221, and 0.3469, respectively, and calculated the difficulty of the question as 0.4832. Next, in the Cognitive Thinking module, MIKT predicted the student's attention

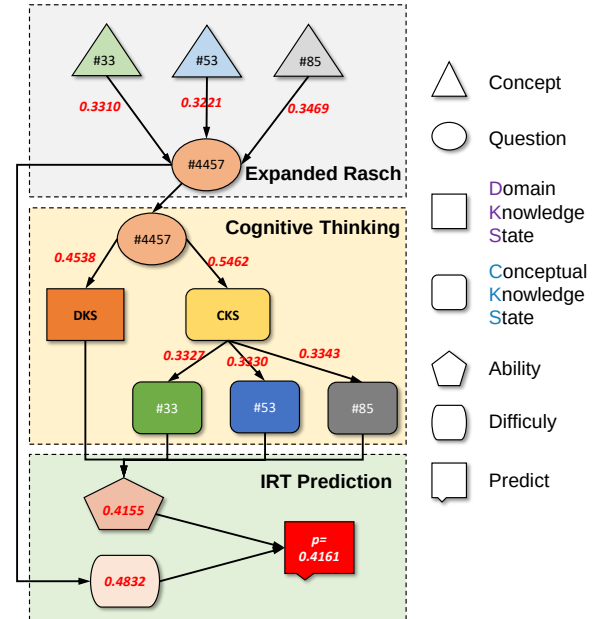


Figure 4: The interpretable prediction process of MIKT.

to domain knowledge state as 0.4538 and the overall attention to conceptual knowledge state as 0.5462. Within conceptual knowledge states, MIKT predicted the student's attention to each relevant conceptual knowledge state as 0.3327, 0.3330, and 0.3343, respectively. Finally, in the IRT prediction module, MIKT combined the student's current knowledge state with the question to calculate the student's ability value for that question as 0.4155. Considering the question's difficulty of 0.4832, MIKT predicted the probability of the student answering the question correctly as 0.4161. Clearly, MIKT's decision process is fully transparent and interpretable. In the context of this example, if the student were to answer the question incorrectly, MIKT could provide the following explanations: (1) The student's ability value is relatively low, which may be due to the student's overall lower proficiency or insufficient mastery of the specific concept/question. (2) The question is relatively challenging for this student. Intuitively, the student's ability value is lower than the question's difficulty, making it difficult for the student to answer correctly. Additionally, MIKT also has the potential to analyze certain student behaviors. For instance, suppose question 4457 is highly correlated with concept 85, and a student demonstrates a

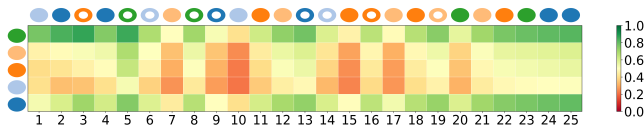
Table 4: Enhance other KT baseline models with MIKT's ERM and compare with Rasch method using AUC.

	DKT	DKT+Rasch	DKT+ERM	DKVMN	DKVMN+Rasch	DKVMN+ERM	SAKT	SAKT+Rasch	SAKT+ERM
ASSIST09	0.7684	0.7827	0.7865	0.7629	0.7751	0.7812	0.7564	0.7715	0.7742
ASSIST12	0.7328	0.7700	0.7719	0.7228	0.7698	0.7690	0.7296	0.7723	0.7665
EdNet	0.7006	0.7626	0.7650	0.6975	0.7608	0.7644	0.6956	0.7626	0.7639
Eedi	0.7629	0.7942	0.7968	0.7590	0.7900	0.7941	0.7556	0.7862	0.7889

strong mastery of concept 85. However, MIKT observes that the student's attention to concept 85 is relatively low. This suggests that, for some reason (such as being unwell and in a suboptimal state), the student may not be able to perform at their true potential. Alternatively, it may indicate that the student's responses are random (i.e., noisy data). We also provide an analysis experiment on the difficulty of the questions in Appendix D.

4.5 Multi-concept Rasch extension (RQ4)

We use the proposed ERM module to enhance some simple and commonly used KT models, including DKT, DKVMN, and SAKT, and compare their performance (AUC) with the Rasch[26] representation method proposed by AKT[10], as shown in Table 4. The ACC results are in Table 7. The approach taken here is the same as AKT's, which means modifying their question representations to ERM's question representation. It can be observed that: (1) Whether using the Rasch method or the ERM method, both significantly improve the performance of these simple KT models. This undoubtedly underscores the importance of question representation. (2) On the ASSIST12 dataset, the ERM method does not consistently outperform the Rasch method. This may be due to the fact that each question in the ASSIST12 dataset is associated with only one concept, and Equation 2 applies a linear transformation to the average related concepts, potentially limiting the expressive power of question variations. (3) In most cases, using the ERM method performs better than using the Rasch method to enhance other models. This indicates the importance of considering questions in relation to multiple related concepts, as opposed to Rasch, which only considers a single concept. The ERM method not only enhances model performance but also aligns better with real-world scenarios.

**Figure 5: Knowledge tracing along a learning sequence.**

5 APPLICATION

The most interesting application of KT may be to trace the knowledge state (the final knowledge state (Equation 7) in this paper) of students. It allows us to gain a better understanding of students' learning progress. In Figure 5, we showed the evolution process of tracing a student's knowledge state using MIKT. Specifically, we randomly selected a student from the ASSIST09 dataset who solved five questions at 25 time steps, we distinguish each question with a different colored circle. Show the corresponding circle

for correct answers, and add a white circle on the respective circle for wrong answers. As shown, when the student answers a question correctly/incorrectly during practice, their mastery of the question clearly strengthens/weakens. Moreover, as time goes by, we find that the student gradually forget questions he/she have previously mastered. In addition, as the number of practice questions increases, the traced knowledge state becomes more stable. Overall, the student's final knowledge state significantly improves compared to the beginning, even though he/she answered some questions incorrectly. This is consistent with our intuition that practicing questions benefits students' knowledge mastery, and even answering questions incorrectly can enhance knowledge.

6 CONCLUSION

In this paper, we propose a novel model called Multiscale-state-based Interpretable Knowledge Tracing (MIKT). It aims to trace students' domain knowledge state and conceptual knowledge state, and the results demonstrate its remarkable effectiveness. Additionally, we extend the Rasch representation method, benefiting not only MIKT but also significantly enhancing the performance of other knowledge tracing baseline models. Furthermore, MIKT introduces IRT approach to improve model interpretability. Experimental results on four commonly used public datasets show that MIKT outperforms current state-of-the-art knowledge tracing models while maintaining high interpretability.

7 LIMITATION AND FUTURE WORK

MIKT brings advantages but also has limitations. Firstly, MIKT lacks the ability to parallelize data processing, which is an inherent drawback based on the state-sequential and state-graph nature of the KT method. Secondly, MIKT struggles with handling noise data in the learning process. Lastly, The question representation learned by the ERM module relies on concepts, eliminating the dependence on concepts in question representation is a highly promising direction.

In terms of future work, we are considering applying the MIKT principles to a Transformer-based KT architecture. Additionally, we aim to incorporate adversarial training to address noise in the data. Taking into consideration the issue of question representation dependency on concepts, we plan to integrate unsupervised learning methods for a more effective approach.

8 ACKNOWLEDGMENTS

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REFERENCES

- [1] Ghodai Abdelrahman and Qing Wang. 2019. Knowledge tracing with sequential key-value memory networks. In *Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval*. Association for Computing Machinery, 175–184.
- [2] Ghodai Abdelrahman and Qing Wang. 2022. Deep graph memory networks for forgetting-robust knowledge tracing. *IEEE Transactions on Knowledge and Data Engineering* (2022).
- [3] Ghodai Abdelrahman, Qing Wang, and Bernardo Nunes. 2023. Knowledge tracing: A survey. *ACM Comput. Surv.* 55, 11 (2023), 1–37.
- [4] Sebastian Bach, Alexander Binder, Grégoire Montavon, Frederick Klauschen, Klaus-Robert Müller, and Wojciech Samek. 2015. On pixel-wise explanations for non-linear classifier decisions by layer-wise relevance propagation. *PLoS one* (2015).
- [5] Jiahao Chen, Zitao Liu, Shuyan Huang, Qiongqiong Liu, and Weiqi Luo. 2023. Improving interpretability of deep sequential knowledge tracing models with question-centric cognitive representations. In *Proceedings of the AAAI Conference on Artificial Intelligence*. AAAI Press.
- [6] Youngduck Choi, Youngnam Lee, Junghyun Cho, Jineon Baek, Byungsoo Kim, Yeongmin Cha, Dongmin Shin, Chan Bae, and Jaewe Heo. 2020. Towards an appropriate query, key, and value computation for knowledge tracing. In *Proceedings of the Seventh ACM Conference on Learning @ Scale*. Association for Computing Machinery, 341–344.
- [7] Youngduck Choi, Youngnam Lee, Dongmin Shin, Junghyun Cho, Seoyon Park, Seewoo Lee, Jineon Baek, Chan Bae, Byungsoo Kim, and Jaewe Heo. 2020. Ednet: A large-scale hierarchical dataset in education. In *International Conference on Artificial Intelligence in Education*. Springer, 69–73.
- [8] Huan Dai, Yue Yun, Yupei Zhang, Wenxin Zhang, and Xuequn Shang. 2022. Contrastive deep knowledge tracing. In *International Conference on Artificial Intelligence in Education*. Springer, 289–292.
- [9] Mingyu Feng, Neil Heffernan, and Kenneth Koedinger. 2009. Addressing the assessment challenge with an online system that tutors as it assesses. *User Modeling and User-Adapted Interaction* 19 (2009), 243–266.
- [10] Aritra Ghosh, Neil Heffernan, and Andrew S Lan. 2020. Context-aware attentive knowledge tracing. In *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. Association for Computing Machinery, 2330–2339.
- [11] Xiaopeng Guo, Zhijie Huang, Jie Gao, Mingyu Shang, Maojing Shu, and Jun Sun. 2021. Enhancing knowledge tracing via adversarial training. In *Proceedings of the 29th ACM International Conference on Multimedia*. Association for Computing Machinery, 367–375.
- [12] Shuyan Huang, Zitao Liu, Xiangyu Zhao, Weiqi Luo, and Jian Weng. 2023. Towards Robust Knowledge Tracing Models via k-Sparse Attention. In *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval*. Association for Computing Machinery, 2441–2445.
- [13] Diederik P Kingma and Jimmy Ba. 2015. Adam: A Method for Stochastic Optimization. In *International Conference on Learning Representations*.
- [14] Jinseok Lee and Dit-Yan Yeung. 2019. Knowledge query network for knowledge tracing: How knowledge interacts with skills. In *LAK19: 9th International Learning Analytics and Knowledge Conference*. 491–500.
- [15] Wonsung Lee, Jaeyoon Chun, Youngmin Lee, Kyoungsoo Park, and Sungrae Park. 2022. Contrastive learning for knowledge tracing. In *Proceedings of the ACM Web Conference 2022*. Association for Computing Machinery, 2330–2338.
- [16] Yunfei Liu, Yang Yang, Xianyu Chen, Jian Shen, Haifeng Zhang, and Yong Yu. 2020. Improving knowledge tracing via pre-training question embeddings. In *Proceedings of the 29th International Joint Conference on Artificial Intelligence*. Morgan Kaufmann, 1577–1583.
- [17] Zitao Liu, Qiongqiong Liu, Jiahao Chen, Shuyan Huang, Boyu Gao, Weiqi Luo, and Jian Weng. 2023. Enhancing deep knowledge tracing with auxiliary tasks. In *Proceedings of the ACM Web Conference 2023*. Association for Computing Machinery, 4178–4187.
- [18] Zitao Liu, Qiongqiong Liu, Jiahao Chen, Shuyan Huang, and Weiqi Luo. 2023. simpleKT: a simple but tough-to-beat baseline for knowledge tracing. In *International Conference on Learning Representations*.
- [19] Yu Lu, Deliang Wang, Qinggang Meng, and Penghe Chen. 2020. Towards interpretable deep learning models for knowledge tracing. In *International Conference on Artificial Intelligence in Education*. Springer, 185–190.
- [20] Koki Nagatani, Qian Zhang, Masahiro Sato, Yan-Ying Chen, Francine Chen, and Tomoko Ohkuma. 2019. Augmenting knowledge tracing by considering forgetting behavior. In *Proceedings of the 28th international conference on World Wide Web*. International World Wide Web Conferences Steering Committee, 3101–3107.
- [21] Hiromi Nakagawa, Yusuke Iwasawa, and Yutaka Matsuo. 2019. Graph-based knowledge tracing: modeling student proficiency using graph neural network. In *IEEE/WIC/ACM International Conference on Web Intelligence*. Association for Computing Machinery, 156–163.
- [22] Qin Ni, Tingjiang Wei, Jiabao Zhao, Liang He, and Chanjin Zheng. 2023. HHSKT: A learner-question interactions based heterogeneous graph neural network model for knowledge tracing. *Expert Systems with Applications* 215 (2023), 119334.
- [23] Shalini Pandey and George Karypis. 2019. A self-attentive model for knowledge tracing. In *Proceedings of the 12th International Conference on Educational Data Mining*. International Educational Data Mining Society, 384–389.
- [24] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, et al. 2019. Pytorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems*. Curran Associates, Inc., 8026–8037.
- [25] Chris Piech, Jonathan Bassen, Jonathan Huang, Surya Ganguli, Mehran Sahami, Leonidas J Guibas, and Jascha Sohl-Dickstein. 2015. Deep knowledge tracing. In *Advances in Neural Information Processing Systems*. Curran Associates, Inc., 505–513.
- [26] Georg Rasch. 1993. *Probabilistic models for some intelligence and attainment tests*. MESA Press.
- [27] Shuanghong Shen, Zhenya Huang, Qi Liu, Yu Su, Shijin Wang, and Enhong Chen. 2022. Assessing Student’s Dynamic Knowledge State by Exploring the Question Difficulty Effect. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*. Association for Computing Machinery, 427–437.
- [28] Shuanghong Shen, Qi Liu, Enhong Chen, Zhenya Huang, Wei Huang, Yu Yin, Yu Su, and Shijin Wang. 2021. Learning process-consistent knowledge tracing. In *Proceedings of the 27th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. Association for Computing Machinery, 1452–1460.
- [29] Xiangyu Song, Jianxin Li, Qi Lei, Wei Zhao, Yunliang Chen, and Ajmal Mian. 2022. Bi-CLKT: Bi-graph contrastive learning based knowledge tracing. *Knowledge-Based Systems* 241 (2022), 108274.
- [30] Xiangyu Song, Jianxin Li, Yifu Tang, Taige Zhao, Yunliang Chen, and Ziyu Guan. 2021. Jkt: A joint graph convolutional network based deep knowledge tracing. *Information Sciences* 580 (2021), 510–523.
- [31] Shashank Sonkar, Andrew E. Waters, Andrew S Lan, Phillip J Grimaldi, and Richard G Baraniuk. 2020. qDKT: Question-centric Deep Knowledge Tracing. In *Proceedings of the 13th International Conference on Educational Data Mining*. International Educational Data Mining Society, 677–681.
- [32] Yu Su, Qingwen Liu, Qi Liu, Zhenya Huang, Yu Yin, Enhong Chen, Chris Ding, Si Wei, and Guoping Hu. 2018. Exercise-enhanced sequential modeling for student performance prediction. In *Proceedings of the AAAI Conference on Artificial Intelligence*. AAAI Press, 2435–2443.
- [33] Jianwen Sun, Shangheng Du, Zhi Liu, Fenghua Yu, Sannyuya Liu, Qing Li, and Xiaoxuan Shen. 2023. Weighted Heterogeneous Graph-based Three-view Contrastive Learning for Knowledge Tracing in Personalized e-Learning Systems. *IEEE Transactions on Consumer Electronics* (2023).
- [34] Jianwen Sun, Fenghua Yu, Sannyuya Liu, Yawei Luo, Ruxia Liang, and Xiaoxuan Shen. 2023. Adversarial bootstrapped question representation learning for knowledge tracing. In *Proceedings of the 31st ACM International Conference on Multimedia*. Association for Computing Machinery, 8016–8025.
- [35] Hanshuang Tong, Zhen Wang, Yun Zhou, Shiwei Tong, Wenyuan Han, and Qi Liu. 2022. Introducing problem schema with hierarchical exercise graph for knowledge tracing. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*. Association for Computing Machinery, 405–415.
- [36] Shiwei Tong, Qi Liu, Wei Huang, Zhenya Huang, Enhong Chen, Chuanren Liu, Haiping Ma, and Shijin Wang. 2020. Structure-based knowledge tracing: An influence propagation view. In *2020 IEEE international conference on data mining (ICDM)*. IEEE, 541–550.
- [37] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. In *Advances in Neural Information Processing Systems*. Curran Associates, Inc., 5998–6008.
- [38] Fei Wang, Qi Liu, Enhong Chen, Zhenya Huang, Yuying Chen, Yu Yin, Zai Huang, and Shijin Wang. 2020. Neural cognitive diagnosis for intelligent education systems. In *Proceedings of the AAAI Conference on Artificial Intelligence*. AAAI Press.
- [39] Zichao Wang, Angus Lamb, Evgeny Saveliev, Pashmina Cameron, Yordan Zaykov, José Miguel Hernández-Lobato, Richard E Turner, Richard G Baraniuk, Craig Barton, Simon Peyton Jones, et al. 2020. Instructions and guide for diagnostic questions: The neurips 2020 education challenge. *arXiv preprint arXiv:2007.12061* (2020).
- [40] Kevin H Wilson, Yan Karklin, Bojian Han, and Chaitanya Ekanadham. 2016. Back to the Basics: Bayesian extensions of IRT outperform neural networks for proficiency estimation. In *Proceedings of the 9th International Conference on Educational Data Mining*. International Educational Data Mining Society, 539–544.
- [41] Tangjie Wu and Qiang Ling. 2023. Self-supervised heterogeneous hypergraph network for knowledge tracing. *Information Sciences* 624 (2023), 200–216.
- [42] Bihan Xu, Zhenya Huang, Jiayu Liu, Shuanghong Shen, Qi Liu, Enhong Chen, Jinze Wu, and Shijin Wang. 2023. Learning Behavior-oriented Knowledge Tracing. In *Proceedings of the 29th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. Association for Computing Machinery, 2789–2800.

- [43] Yang Yang, Jian Shen, Yanru Qu, Yunfei Liu, Kerong Wang, Yaoming Zhu, Weinan Zhang, and Yong Yu. 2021. GIKT: a graph-based interaction model for knowledge tracing. In *Machine Learning and Knowledge Discovery in Databases*. Springer, 299–315.
- [44] Chun-Kit Yeung. 2019. Deep-IRT: Make deep learning based knowledge tracing explainable using item response theory. In *Proceedings of the 12th International Conference on Educational Data Mining*. International Educational Data Mining Society, 683–686.
- [45] Chun-Kit Yeung and Dit-Yan Yeung. 2018. Addressing two problems in deep knowledge tracing via prediction-consistent regularization. In *Proceedings of the fifth annual ACM conference on learning at scale*. 1–10.
- [46] Yu Yin, Le Dai, Zhenya Huang, Shuanghong Shen, Fei Wang, Qi Liu, Enhong Chen, and Xin Li. 2023. Tracing Knowledge Instead of Patterns: Stable Knowledge Tracing with Diagnostic Transformer. In *Proceedings of the ACM Web Conference 2023*. Association for Computing Machinery, 855–864.
- [47] Jiani Zhang, Xingjian Shi, Irwin King, and Dit-Yan Yeung. 2017. Dynamic key-value memory networks for knowledge tracing. In *Proceedings of the 26th International Conference on World Wide Web*. International World Wide Web Conferences Steering Committee, 765–774.

A ONE-PARAMETER IRT

The primary model utilized in this paper is the one-parameter IRT model, and its mathematical formula is presented below:

$$P(X_{ij} = 1) = \sigma(\theta_i - \beta_j) \quad (15)$$

Among them, $P(X_{ij} = 1)$ represents the probability of student i correctly answering question j , σ represents the sigmoid function, θ_i represents the ability value of student i for this question, and β_j represents the difficulty value of question j .

B DATASETS

We evaluated the performance of MIKT on four commonly used public datasets:

- **ASSIST09**[9]: Collected from the ASSISTments online educational platform during 2009-2010.
- **ASSIST12**[9]: Collected from the ASSISTments online educational platform during 2012-2013.
- **EdNet**[7]: A dataset collected by Santa[7], an online tutoring platform, from 2017 to 2019.
- **Eedi**[39]: Used for the NeurIPS 2020 Education Data Mining Challenge, collected by the online education platform Eedi from 2018 to 2020.

Based on previous research, for the ASSIST series datasets, we removed scaffold questions and records without concepts[10]. Moreover, due to the large scale of ASSIST12, EdNet, and Eedi datasets, and limitations in computational resources, we randomly selected 5,000 students uniformly without duplication[43]. The statistical information for these datasets is provided in Table 1.

C BASELINE

To assess the performance of MIKT, we compared it with 20 state-of-the-art KT models as follows:

- **DKT**[25]: Traces students' knowledge states using LSTM.
- **DKT+**[45]: Enhances DKT by addressing inconsistent knowledge states and irrecoverable inputs.
- **KQN**[14]: Predicts students' performance using knowledge state encoder and concept encoder.
- **DKT+forgetting**[20]: Augments DKT by incorporating various behavioral features to consider forgetting in student knowledge states.
- **PEBG+DKT**[16]: Enhances DKT by deeply exploring the relationship between questions and concepts to obtain pre-trained question representations.
- **GIKT**[43]: Use GCN to aggregate the relationship between questions and concepts to enhance question representation.
- **ATKT**[11]: Use adversarial training to improve the generalization ability of the model.
- **CDKT**[8]: On the basis of DKT, use contrastive learning between questions to represent questions.

- **DIMKT**[27]: Traces the impact of question difficulty on students' knowledge states.
- **QIKT**[5]: Models centered around questions and enhances model interpretability with interpretable parameters.
- **AT-DKT**[17]: Enhances DKT with two additional tasks to improve performance.
- **DKVMN**[47]: Traces students' knowledge states using a dynamic key-value memory network.
- **Deep-IRT**[44]: Integrates IRT with DKVMN to improve interpretability.
- **GKT**[21]: Propagates students' conceptual knowledge states using a graph structure.
- **SAKT**[23]: Captures student-concept relationships using self-attention mechanism.
- **SAINT**[6]: Fully employs a Transformer[37] architecture to model students' knowledge states.
- **AKT**[10]: Simulates students' forgetting behavior using context-based attention mechanism.
- **CLAKT**[15]: Mitigates sparsity in student-concept interactions using contrastive learning.
- **simpleKT**[18]: Simplifies the model structure based on AKT, achieving simplicity without sacrificing performance.
- **DTransformer**[46]: Uses contrastive learning to trace students' stable knowledge states.

It is worth noting that all the KT models compared to MIKT share the same input configuration, which includes only the question, concept, and response as input features (this is also the most common input setup in KT research). KT models, such as DGMN[2], SKT[36], LPKT[28], and LBKT[42], that require additional input features were not included in the comparison because it could lead to an unfair performance comparison.

Table 5: The Pearson correlation coefficient between the difficulty learned by MIKT and the statistical difficulty.

Dataset	ASSIST09	ASSIST12	EdNet	Eedi
Pearson coefficient	0.7523	0.7406	0.8964	0.7728

D QUESTION DIFFICULTY ANALYSIS

In this section, we will analyze the validity of the difficulty learned by MIKT. Specifically, we will compare the learned difficulty values (denoted as $diff_{q_i}$ in Equation 2) with the statistical difficulty (i.e., the question error rates in the dataset). We utilize the Pearson correlation coefficient as the evaluation metric, which ranges from -1 to 1, with values closer to 1 indicating a stronger correlation. The results are presented in Table 5, where we observe a very high correlation between the difficulty values learned by MIKT and the statistical difficulty (mostly exceeding 0.75). This highlights the effectiveness of the difficulty learned by MIKT.

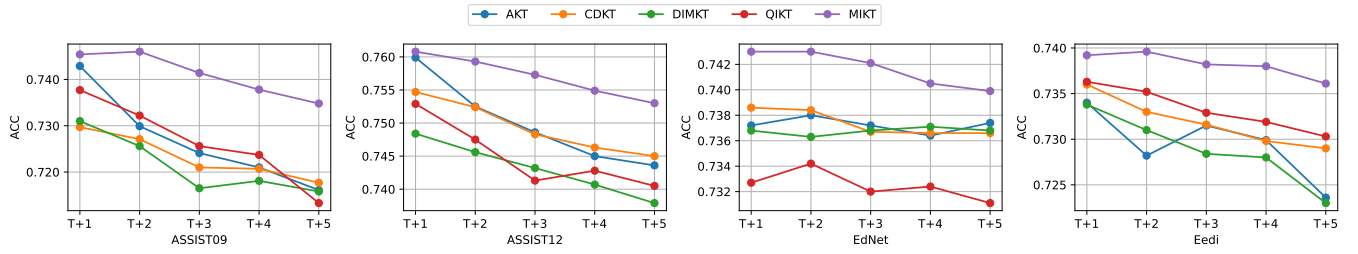


Figure 6: Comparing T+N prediction performance (ACC) of MIKT and well-performing KT models across four datasets.

Table 6: The impact of the ERM module on various methods. "gap" indicates how much the model's performance (AUC) decreased after removing the ERM module.

	DKT+ERM	DKT	gap	DKVMN+ERM	DKVMN	gap	SAKT+ERM	SAKT	gap	MIKT	MIKT w/o ERM	gap
ASSIST09	0.7865	0.7684	-1.81%	0.7812	0.7629	-1.83%	0.7742	0.7564	-1.78%	0.7938	0.7822	-1.16%
ASSIST12	0.7719	0.7328	-3.91%	0.7690	0.7228	-4.62%	0.7665	0.7296	-3.69%	0.7834	0.7787	-0.47%
EdNet	0.7650	0.7006	-6.44%	0.7644	0.6975	-6.69%	0.7639	0.6956	-6.83%	0.7703	0.7669	-0.34%
Eedi	0.7968	0.7629	-3.39%	0.7941	0.7590	-3.51%	0.7889	0.7556	-3.33%	0.7954	0.7925	-0.29%

Table 7: Enhance other KT baseline models with MIKT's ERM and compare with Rasch method using ACC.

	DKT	DKT+Rasch	DKT+ERM	DKVMN	DKVMN+Rasch	DKVMN+ERM	SAKT	SAKT+Rasch	SAKT+ERM
ASSIST09	0.7297	0.7381	0.7393	0.7266	0.7322	0.7375	0.7192	0.7281	0.7297
ASSIST12	0.7367	0.7520	0.7540	0.7329	0.7534	0.7523	0.7348	0.7542	0.7512
EdNet	0.7129	0.7380	0.7397	0.7120	0.7367	0.7382	0.7115	0.7370	0.7377
Eedi	0.7182	0.7380	0.7387	0.7162	0.7346	0.7369	0.7123	0.7318	0.7329